

SQL Views Practice — Questions

Database: COMPANY by Eng | Ahmed Salem

Part 1 — Simple Views

Q1

Create a view named `employee_basic_view` that displays:

- `fname`
- `lname`
- `salary`

From the employee table.

Q2

Create a view named `research_employees` that displays all employees who work in department number 5.

Q3

Create a view named `high_salary_employees` that displays: for employees whose salary is greater than 30000:

- `fname`
- `lname`
- `salary`

Q4

Create a view named `female_employees` that displays: for female employees only:

- `fname`
- `lname`
- `sex`
- `salary`

Q5

Create a view named `employee_names` that displays:

- `ssn`
- `full name`
- `salary`

The full name should contain `fname`, `minit`, and `lname`.

Q6

Create a view named `department5_employees` that displays: for employees working in department 5:

- `ssn`
- `fname`
- `lname`
- `salary`
- `dno`

Q7

Create a view named `employee_addresses` that displays:
for all employees:

- `fname`
- `lname`
- `address`

Q8

Create a view named `low_salary_employees`
that displays employees whose salary is less than or equal to 25000.

Part 2 — DML Through Simple Views

Q9

Create a view containing:

- `fname`
- `lname`
- `salary`

Then use the view to increase John Smith's salary to 35000.

Q10

Create a view containing employees who work in department number 4.

Then use the view to change Alicia Zelaya's salary to 28000.

Q11

Create a view containing:

- `fname`
- `lname`
- `salary`

Then delete Ahmad Jabbar through the view.

Question: Does the DELETE affect the original employee table?

Part 3 — Complex Views: JOIN

Q12

Create a view named `employee_department_view` that displays:

- `employee name`
- `department name`

Use employee and department tables.

Q13

Create a view named `employee_projects` that displays:

- `fname`
- `lname`
- `pname`

Show every employee and the projects they work on.

You will need: `employee`, `works_on`, `project`

Q14

Create a view named `project_department_view` that displays:

- `pname`
- `plocation`
- `dname`

Show each project and the department responsible for it.

Q15

Create a view named `employee_project_hours` that displays:

- employee name
- project name
- hours

Example: John Smith | ProductX | 32.5

Part 4 — Complex Views: Aggregate Functions

Q16

Create a view named `department_avg_salary` that displays:

- `dno`
- `average_salary`

Calculate the average salary for each department.

Use: `AVG()`, `GROUP BY`

Q17

Create a view named `department_employee_count` that displays:

- `dno`
- `employee_count`

Calculate the number of employees in each department.

Use: `COUNT()`, `GROUP BY`

Q18

Create a view named `department_total_salary` that displays:

- `dno`
- `total_salary`

Calculate the total salary for each department.

Use: `SUM()`, `GROUP BY`

Q19

Create a view named `department_max_salary` that displays:

- `dno`
- `max_salary`

Find the highest salary in each department.

Use: MAX(), GROUP BY

Part 5 — Complex Views: JOIN + GROUP BY

Q20

Create a view named `department_info` that displays:

- `dname`
- `employee_count`
- `average_salary`

For each department show: department name, number of employees, average salary.

You will need: employee, department, JOIN, COUNT(), AVG(), GROUP BY

Q21

Create a view named `project_hours_summary` that displays:

- `pname`
- `total_hours`

Calculate the total number of hours spent on each project.

You will need: project, works_on, SUM(), GROUP BY

Q22

Create a view named `employee_total_hours` that displays:

- `fname`
- `lname`
- `total_hours`

Calculate the total hours worked by each employee on all projects.

Use: SUM(), GROUP BY

Q23

Create a view named `department_project_count` that displays:

- `dname`
- `project_count`

Calculate the number of projects for each department.

Part 6 — More Complex Views

Q24

Create a view that displays:

- employee name
- department name
- project name
- hours

Example: John Smith | Research | ProductX | 32.5

You will need multiple tables.

Q25

Create a view named `research_project_hours` that displays:

- project name
- total hours

Show only projects belonging to department 5. Calculate total hours for each project.

Q26

Create a view that displays the department with the highest average salary.

Use aggregate functions.

Q27

Create a view that displays employees who work on more than one project.

Think about: COUNT(), GROUP BY, HAVING

Q28

Create a view that displays employees whose total project hours are greater than 30.

- employee_name
- total_hours

Think about: SUM(), GROUP BY, HAVING

Part 7 — Simple or Complex?

Q29 — Is this view SIMPLE or COMPLEX? Explain why.

```
CREATE VIEW v1 AS
SELECT fname, lname, salary
FROM employee;
```

Q30 — Is this view SIMPLE or COMPLEX? Explain why.

```
CREATE VIEW v2 AS
SELECT e.fname, d.dname
FROM employee e
JOIN department d
  ON e.dno = d.dnumber;
```

Q31 — Is this view SIMPLE or COMPLEX? Explain why.

```
CREATE VIEW v3 AS
SELECT dno, AVG(salary)
FROM employee
GROUP BY dno;
```

Q32 — Is this view SIMPLE or COMPLEX? Can we perform UPDATE through this view?

```
CREATE VIEW v4 AS
SELECT *
FROM employee
WHERE salary > 30000;
```

Q33 — Is this view SIMPLE or COMPLEX? Explain why.

```
CREATE VIEW v5 AS
SELECT dname, COUNT(*)
FROM department d
JOIN employee e
  ON d.dnumber = e.dno
GROUP BY dname;
```

Part 8 — Final Challenge

Try to create the following views without looking at any solution:

- simple_employee_view
- research_employee_view
- employee_department_view
- employee_project_view
- department_avg_salary
- department_employee_count
- employee_total_hours
- project_total_hours
- department_project_count
- employees_more_than_one_project

Important — Simple vs Complex View

Simple View	Complex View
Usually based on ONE table	May use multiple tables
No GROUP BY	May use GROUP BY
No aggregate functions	May use aggregate functions
DML is generally possible	DML may not be possible